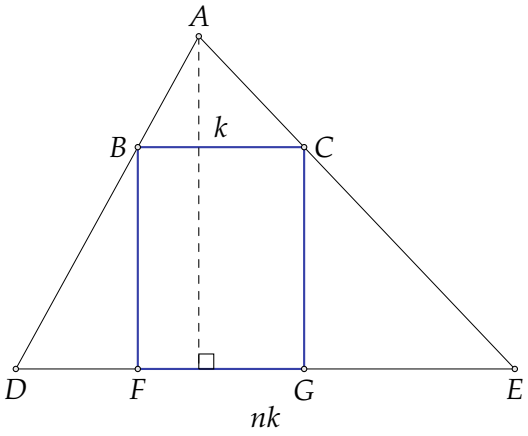
Cayley Olympiad 2023, Problem 4: four scripts, four marks

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The problem

Rectangle $BCGF$ lies inside triangle ADE, with B on AD, C on AE, and F, G on DE. The base DE is n times the length of BC, the shorter side of the rectangle. Prove that the rectangle occupies a fraction $\frac{2}{n} - \frac{2}{n^2}$ of the area of the triangle.



Write $BC = k$, so $DE = nk$, and let h_1, h_2 be the heights of triangles ABC and ADE from A.

Script A **10/10**

Let the shorter length of the rectangle be k; thus the base of the triangle has a length of nk. Since $BCGF$ is a rectangle, BC is parallel to FG, which is on DE. So BC is also parallel to DE. So $\angle ABC = \angle ADE$ and $\angle ACB = \angle AED$. Also $\angle BAC = \angle DAE$ (share the same angle). So triangles ABC and ADE are similar. Let the height of $\triangle ABC$ be h_1 and the height of $\triangle ADE$ be h_2. So

$$\frac{BC}{DE} = \frac{k}{nk} = \frac{1}{n} = \frac{h_1}{h_2},$$

so $h_2 = nh_1$. So the height of the rectangle is $h_2 - h_1 = nh_1 - h_1 = (n-1)h_1$. Therefore the area of the rectangle is $k \times (n-1)h_1$, and the area of triangle ADE is $nk \times nh_1 \times \frac{1}{2} = \frac{n^2 k h_1}{2}$. So the fraction of the total area that the rectangle occupies is

$$\frac{k(n-1)h_1}{\frac{1}{2}n^2 k h_1} = \frac{2(n-1)}{n^2} = \frac{2n-2}{n^2} = \frac{2}{n} - \frac{2}{n^2}.$$

This question is therefore proved.

Marker's comment. A complete proof. The parallel sides come from the rectangle, the similarity from equal angles, and the heights are named before they are used. Two facts go unstated, that heights of similar triangles scale with the sides and that the rectangle's height is $h_2 - h_1$ because $BC \parallel DE$, but no marker asks for these at Cayley level.

Script B 6/10

Triangle ABC is clearly similar to triangle ADE with scale factor n, so the height of ADE is n times the height h of ABC. The rectangle has width k and height $(n-1)h$, and the triangle has area $\frac{1}{2} \cdot nk \cdot nh$. Dividing gives $\dfrac{2(n-1)}{n^2} = \dfrac{2}{n} - \dfrac{2}{n^2}$.

Marker's comment. The calculation is right, but the one step that needs proof is called "clear". The script should say that $BC \parallel DE$ as opposite sides of the rectangle, so corresponding angles are equal and the triangles are similar. Asserting the main step costs more than a careless slip would.

Script C 2/10

Triangles ABC and ADE are similar because $BC \parallel DE$, with scale factor n, so the area of ADE is n^2 times the area of ABC. The rectangle is the part of triangle ADE outside triangle ABC, so the rectangle occupies $\dfrac{n^2-1}{n^2}$ *of the triangle, which equals* $\dfrac{2}{n} - \dfrac{2}{n^2}$.

Marker's comment. The similarity and the area ratio n^2 earn the marks. The part of triangle ADE outside triangle ABC also contains triangles BDF and CGE, so the region is not the rectangle. The last step then claims $\frac{n^2-1}{n^2} = \frac{2}{n} - \frac{2}{n^2}$, which fails at $n = 2$, and forcing the given answer after a false step loses everything after that step.

Script D 1/10

Take $n = 3$, $DE = 6$ and a height of 4, so $BC = 2$. By similar triangles BC sits at height $\frac{8}{3}$, so the rectangle has area $2 \times \frac{8}{3} = \frac{16}{3}$ and the triangle has area 12. The fraction is $\frac{16/3}{12} = \frac{4}{9} = \frac{2}{3} - \frac{2}{9}$, as required.

Marker's comment. Correct for one triangle and one value of n, which proves nothing in general. At most one mark, for understanding the configuration. The same working with nk, h_2 and n in place of 6, 4 and 3 is Script A.

What full marks needs

The proof needs $BC \parallel DE$ from the rectangle, the similarity from equal angles, and the rectangle's height as $h_2 - h_1$. Script B asserts the first two, and Script C replaces the third with the wrong region.